

EXPERIMENT 15

Determination and plotting of Inductance (i) Self-inductance and Mutual inductance of current carrying coil, (ii) solenoid and toroid with different (a) length and (b) circumferential area.

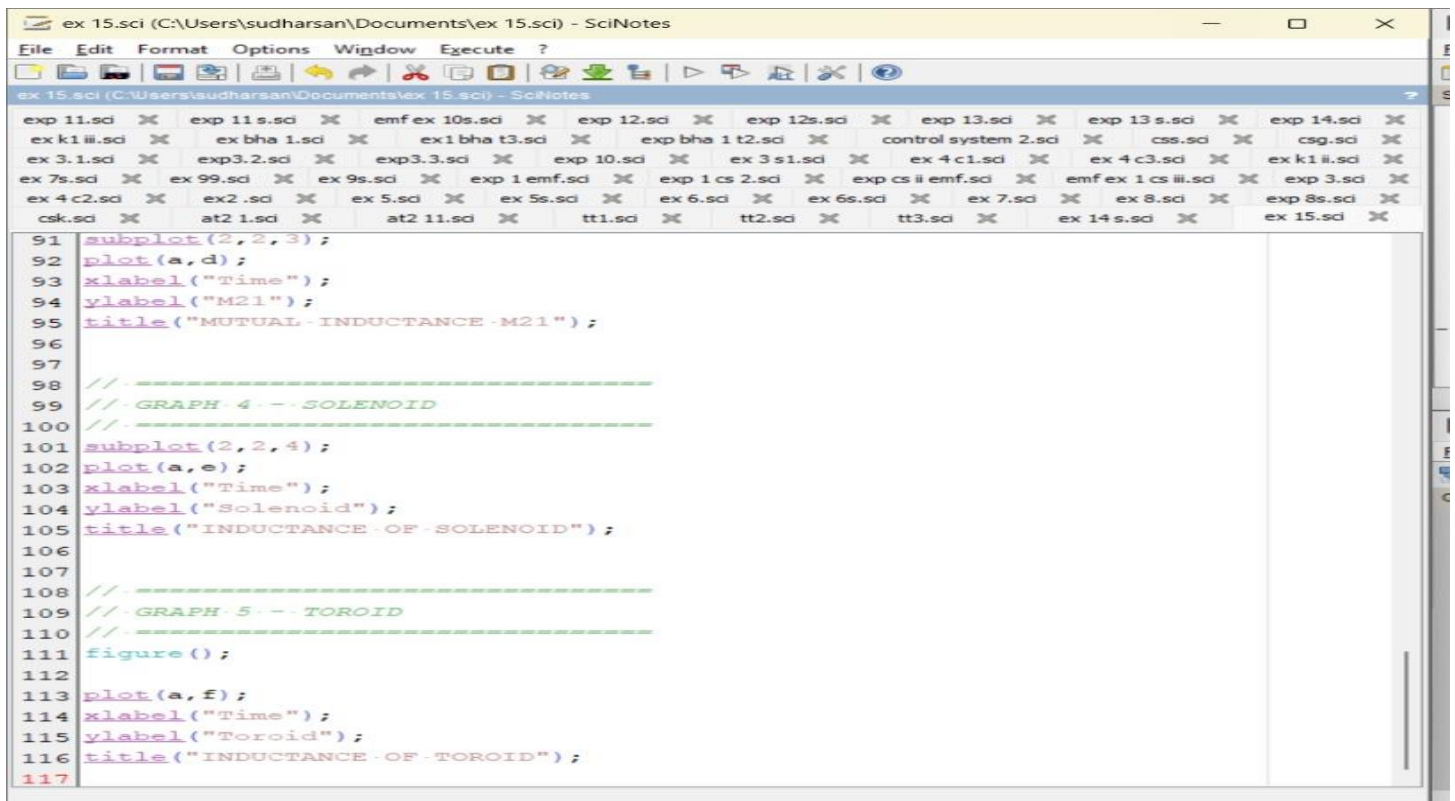
Aim:

To determine and plot Inductance (i) Self-inductance and Mutual inductance of current carrying coil, (ii) solenoid and toroid with different (a) length and (b) circumferential area.

Software required:

SCILAB version 6.0.1.

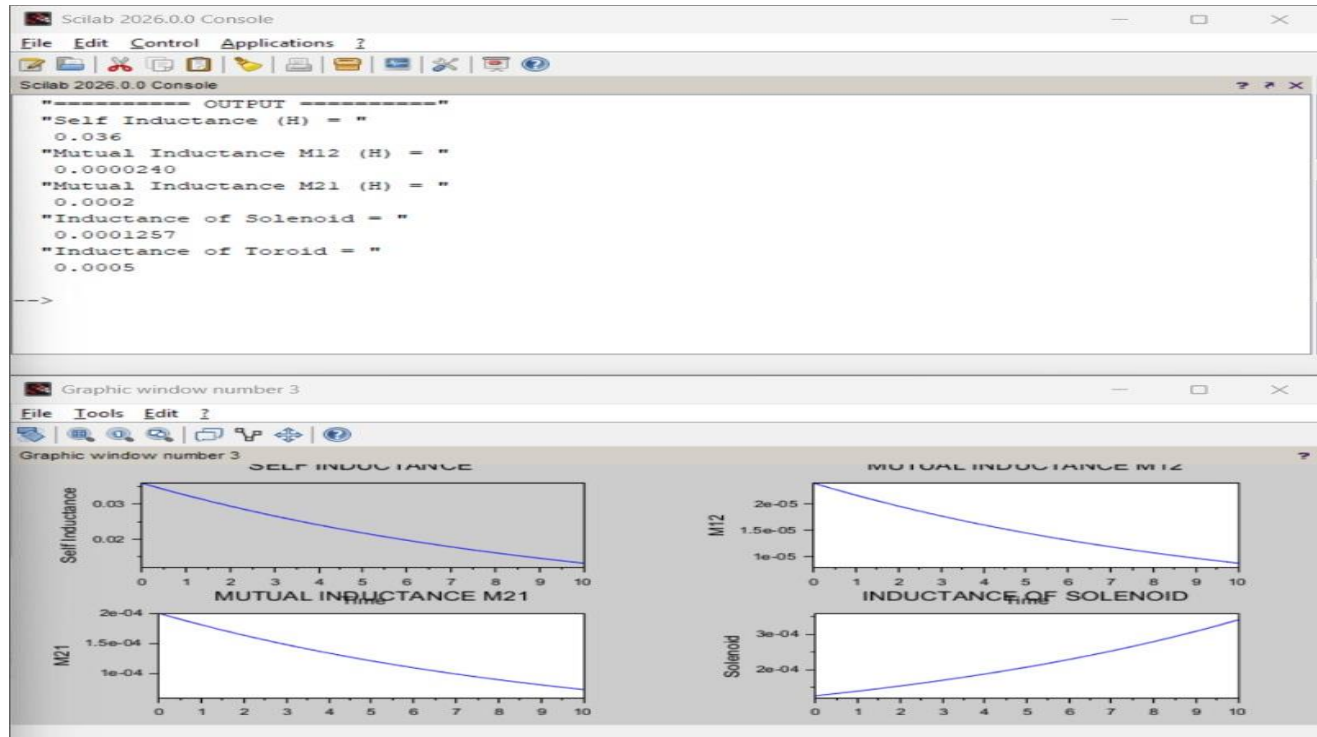
Program:



```
ex 15.sci (C:\Users\sudharsan\Documents\ex 15.sci) - SciNotes
File Edit Format Options Window Execute ?
exp 11.sci X exp 11.s.sci X emfex 10s.sci X exp 12.sci X exp 12s.sci X exp 13.sci X exp 13.s.sci X exp 14.sci X
ex k1ii.sci X ex bha 1.sci X ex 1 bha t3.sci X exp bha 1 t2.sci X control system 2.sci X css.sci X csg.sci X
ex 3.1.sci X exp3.2.sci X exp3.3.sci X exp 10.sci X ex 3 s1.sci X ex 4 c1.sci X ex 4 c3.sci X ex k1ii.sci X
ex 7s.sci X ex 99.sci X ex 9s.sci X exp 1 emf.sci X exp 1 cs 2.sci X exp cs ii emf.sci X emfex 1 cs iii.sci X exp 3.sci X
ex 4c2.sci X ex 2.sci X ex 5.sci X ex 5s.sci X ex 6.sci X ex 6s.sci X ex 7.sci X ex 8.sci X exp 8s.sci X
csk.sci X at2 1.sci X at2 11.sci X tt1.sci X tt2.sci X tt3.sci X ex 14.s.sci X ex 15.sci X

91 subplot(2,2,3);
92 plot(a,d);
93 xlabel("Time");
94 ylabel("M21");
95 title("MUTUAL - INDUCTANCE - M21");
96
97
98 //-----
99 //--GRAPH 4--SOLENOID
100 //-----
101 subplot(2,2,4);
102 plot(a,e);
103 xlabel("Time");
104 ylabel("Solenoid");
105 title("INDUCTANCE - OF - SOLENOID");
106
107
108 //-----
109 //--GRAPH 5--TOROID
110 //-----
111 figure();
112
113 plot(a,f);
114 xlabel("Time");
115 ylabel("Toroid");
116 title("INDUCTANCE - OF - TOROID");
117
```

Output:



Theoretical calculations:

Exp: 15

$$I_1 = 5A$$

$$I_2 = 6A$$

$$l = 5cm$$

$$r_1 = 2mm$$

$$L = \frac{N\Phi}{I} = \frac{6(3 \times 10^{-3})}{5} = 0.0364$$

$$\mu_{12} = \frac{N_2 \Phi_{12}}{I_1} = \frac{600(2 \times 10^{-4})}{5} = 2.4 \times 10^{-5}$$

$$\mu_{21} = \frac{N_1 \Phi_{21}}{I_2} = \frac{400(3 \times 10^{-6})}{6} = 2 \times 10^{-5}$$

$$= \frac{4 \times 10^{-7} \times i}{\text{length}} = \frac{4\pi \times 10^{-7} \times 5}{0.05}$$

$$= 1.25 \times 10^{-4}$$

Result:

Thus determination and plotting of Inductance ,Self-inductance, Mutual inductance of current carrying coil, solenoid and toroid with different (a) length and (b) circumferential area are performed and verified successfully.